Stack Overflow in the AI Era: Review 2 final report

Team: Abhinav (M1), Rishiikesh (M2), Akshay (M3), Nishanth (M4), Adhikkesh (M5). **Scope:** RQ4 (topics and complexity) and RQ5 (technology forecasting). All numbers are generated from `reports/review2/` by `src/review2_final_report.py`.

1. Executive summary

- **Patterns and the 2022 break.** Trend dominates every series. On the log scale, all 14 series show a significant negative change in weekly growth after 2022-11-30 (ITS slope change -3.89% to -1.60% per week, 95% CI excluding zero), and the real break exceeds the 2021-11-30 placebo in 13 of 14 series.
- **Forecasts.** The 26-week site total is forecast to fall from 215 to 134 questions per week by February 2027 (-37.6%), with 6 tags classed steep decline and 5 declining under the fixed thresholds.
- **Topics.** NMF k=10 wins the pre-registered rule (coherence 0.2175, diversity 0.80, stability 1.00). Routine programming topics lose share while harder runtime and tooling topics gain: Python Execution & Runtime Errors rises +1.9 pp while Core Syntax, Arrays & Data Structures falls -3.5 pp.
- **Tone and complexity.** Questions are longer (+13 prose words) and more code-heavy (+4 code lines) after the break. Tone measures are exploratory: the frustration lexicon moves only -0.2 pp.
- **Recommendations.** Six combined recommendations follow, each citing one time-series and one text-mining number; the headline is to move moderation effort to the steep-decline tags and to route rising low-resolution topics to experts.

2. Data recap

- **Volume series.** 302,026 cleaned questions back a weekly series of 14 groups (ALL plus 13 study tags) x 400 complete Tuesday-start weeks, 2019-01-01 to 2026-08-25, with no missing weeks. The time-series method uses these complete counts, never the sampled records.
- **Rate series.** `data/processed/monthly_tag_tables.csv` holds the weighted monthly resolved rate, answered rate and median hours per tag. Months flagged `low_sample` are excluded from conclusions.
- **Text.** 302,026 questions; 165,887 distinct terms after cleaning; code removed from the prose field.
- **Weighting.** Before 2025 every question carries a sampling weight; any share, rate or count built before 2025 is weighted. From 2025 the weight is 1.
- **Periods.** Pre-AI 2019-01 to 2022-10, transition 2022-11 to 2022-12, post-AI 2023-01 to 2026-08. Event markers: GitHub Copilot GA (2022-06-21); ChatGPT Launch (2022-11-30); Stack Overflow ChatGPT Ban (2022-12-05); Stack Overflow Moderator Strike (2023-06-05).

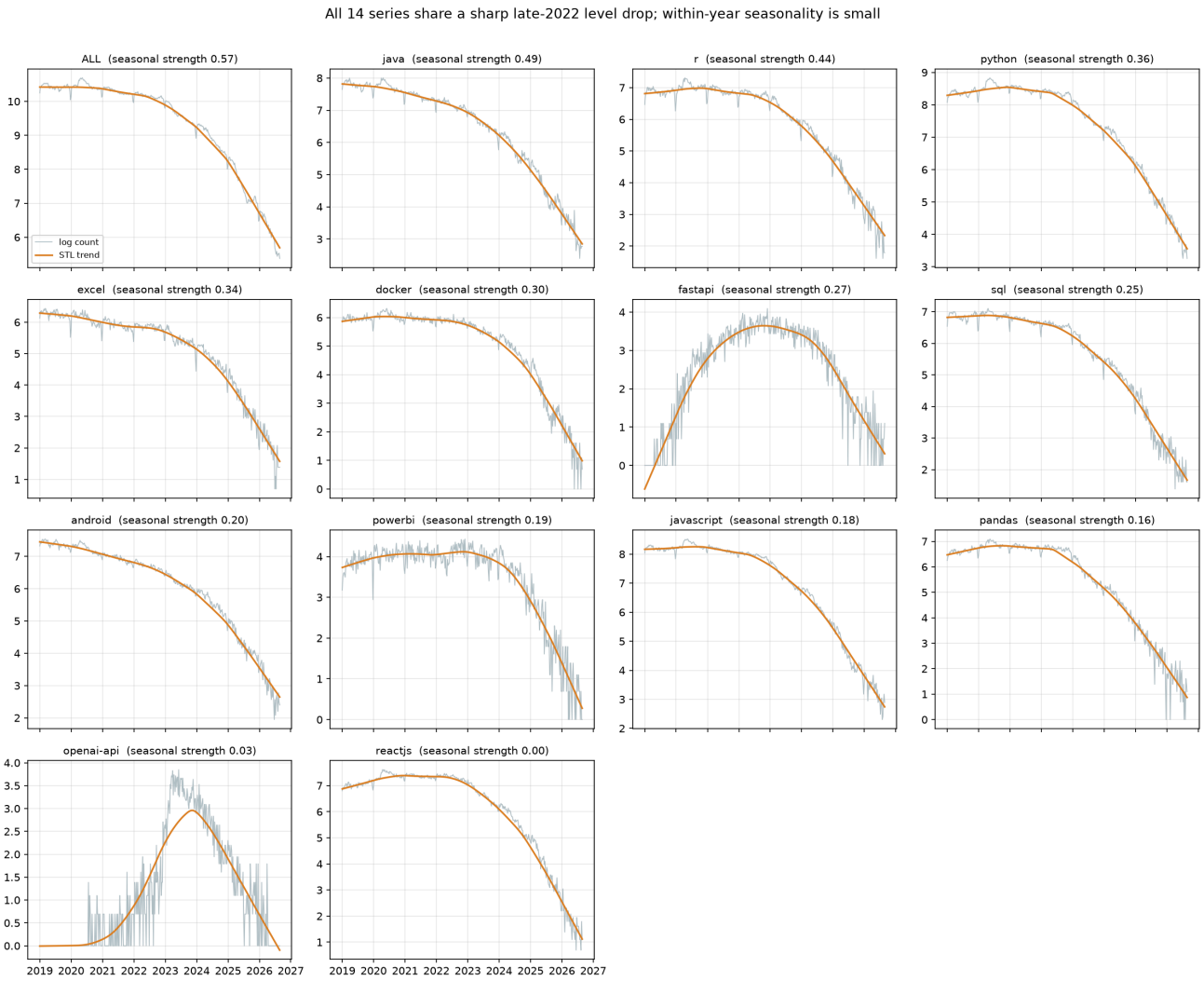
3. Method 1: time series

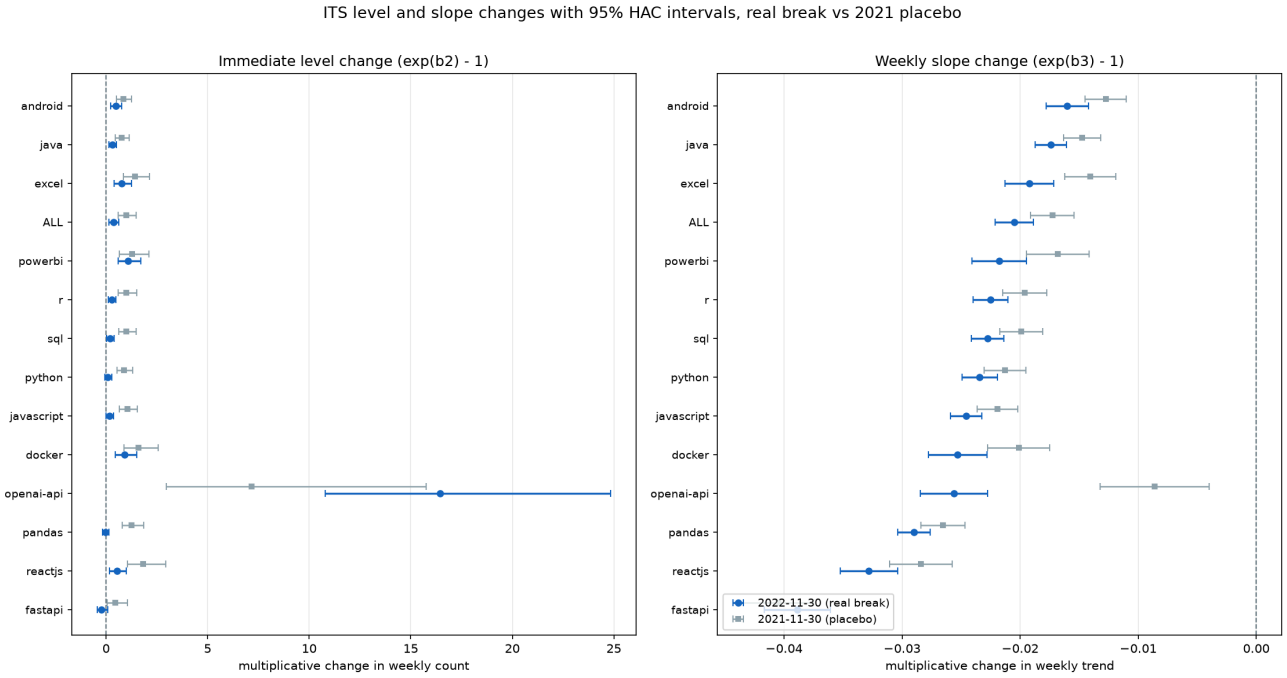
Preparation and implementation. STL (robust, period 52) was fit to each log1p weekly count; the additive components satisfy trend + seasonal + residual = log count. Stationarity was tested with ADF (regression c, AIC) and KPSS on the level and the first difference, under a written verdict rule. Change points were found with `ruptures` PELT (l2 cost, minimum segment 26 weeks) and a BIC penalty fixed in advance, then compared with the four dated events. Interrupted time series were fit per series as segmented regressions of log count on week, a step at 2022-11-30 and a post-break slope, with two Fourier pairs, using OLS with Newey-West HAC errors (lag 4) and a 2021-11-30 placebo.

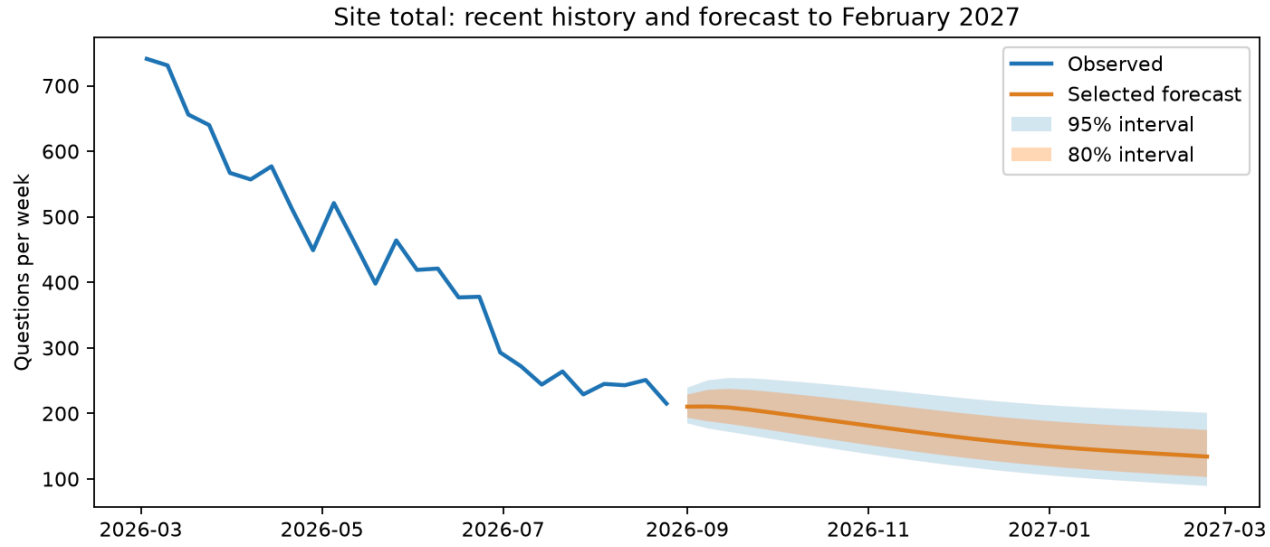
Evaluation and interpretation.

- **Seasonality is secondary.** Trend strength runs 0.86 to 1.00; seasonal strength runs 0.00 to 0.57. Yearly dips cluster in the late-December/January holiday weeks.

- **One difference is justified.** All 14 level series are non-stationary (ADF cannot reject a unit root, KPSS rejects stationarity), so every series is first-differenced (`d=1`), consistent with the M5 SARIMAX backtest. The first-difference tests conflict for 12 series and are reported as such, not hidden.
- **A genuine 2022 break.** The 134 detected breaks are mostly routine noise: 6 fall within four weeks of a dated event, across 4 series (`java`, `r`, `openai-api`, `sql`). The interrupted time series is sharper: all 14 series show a significant negative post-break slope, -3.89% to -1.60% per week, and the real effect exceeds the placebo in 13 of 14 series. 10 series also show a significant immediate level change, most extreme for `openai-api` (+1645%), which is a step, not a stable trend.
- **Forecast skill is earned at long horizons, not short.** On the 2025-26 rolling-origin backtest, SARIMAX's mean 26-week skill against naive is 55.9% while true 95% coverage is 52.0% (intervals too narrow). At one week ahead the best model reaches 15.7% skill and drift only 0.4%, so the naive baseline remains hard to beat over a single week, as expected.
- **Classes and business reading.** Under the fixed thresholds (5 declining, 2 not classifiable: zero baseline, 6 steep decline), the site total is forecast from 215 to 134 questions per week by February 2027 (-37.6%). `tag_classes.csv` adds a half-life and a 2025 index; zero-baseline tags cannot support a percentage class and are labelled as such.







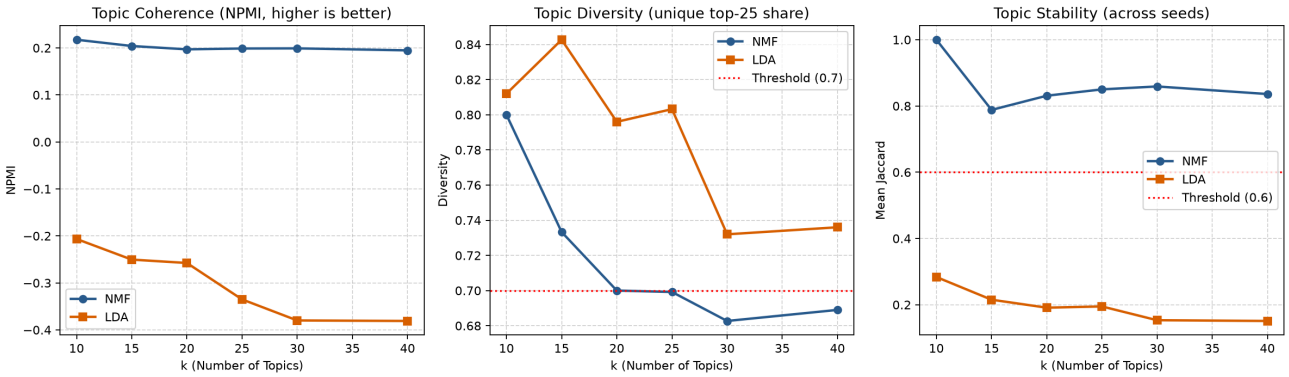
4. Method 2: text mining

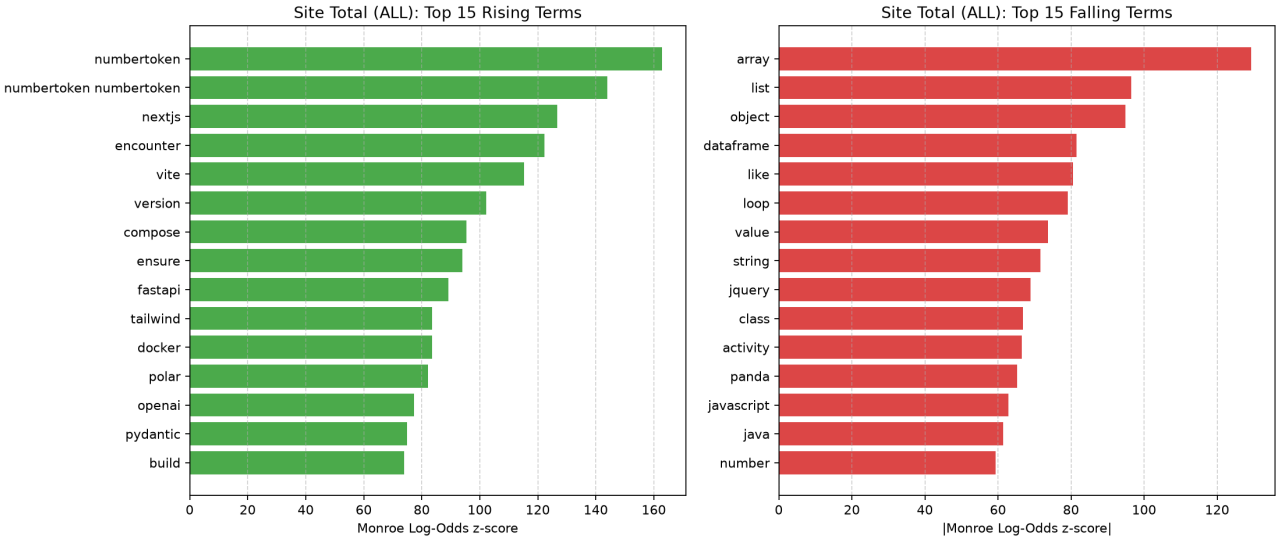
Preparation and implementation. The corpus of 302,026 questions was lowercased, lemmatised with spaCy and stripped of code, with technical words protected and stopwords and VADER overrides written down. Topics were fit by TF-IDF (unigrams and bigrams) on a stratified sample, with NMF and LDA compared across k under a rule fixed in advance (highest coherence among candidates with diversity at least 0.7 and stability at least 0.6). Tone used VADER plus a frustration and politeness lexicon; complexity measured prose length, code lines, code blocks, reading ease and structural flags. Shares and rates before 2025 are weighted.

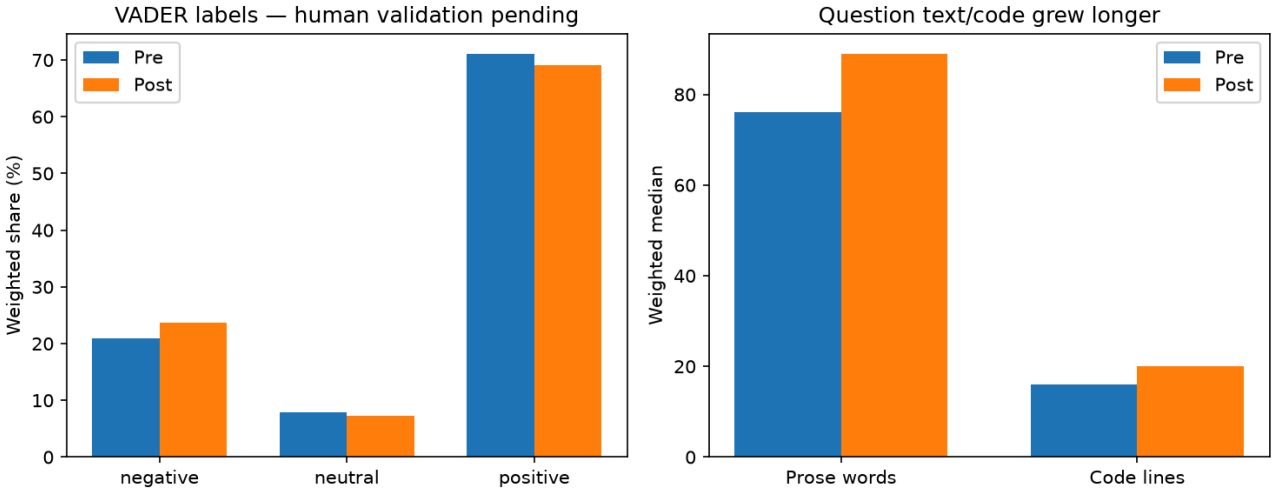
Evaluation and interpretation.

- **NMF k=10 wins.** Coherence 0.2175, diversity 0.80, stability 1.00. LDA failed the rule on every k (coherence -0.381 to -0.206, stability at or below 0.28).
- **What moved to AI tools.** The largest falling topics are Core Syntax, Arrays & Data Structures (-3.5 pp), Pandas Dataframe & Data Manipulation (-1.1 pp), Web Frontend, DOM & User Events (-0.5 pp): routine syntax and dataframe lookups.

- **What people still bring to humans.** The largest rising topics are Python Execution & Runtime Errors (+1.9 pp), Docker Containers & Cloud Deployment (+1.5 pp), Android Mobile App & Gradle Build (+1.1 pp); the top rising topic resolves only 33.2% within seven days, the resolution bottleneck behind the expert-routing recommendation.
- **Vocabulary shift.** Rising terms include nextjs (z=127), encounter (z=122), vite (z=115), version (z=102), compose (z=95); falling terms include array (z=-129), list (z=-97), object (z=-95), dataframe (z=-82), like (z=-81) (Monroe log-odds with a Dirichlet prior).
- **Tone and complexity.** Negative tone share changed +2.7 pp and positive share -2.0 pp; the frustration flag changed -0.2 pp and is a lexicon measure, not verified emotion. Weighted median prose length changed +13 words and code lines +4, while questions spanning two or more study tags changed -3.4 pp.







Tone validation is **pending**: 0 of 200 questions are independently hand-labelled, so accuracy, macro-F1 and Cohen's kappa are not claimed. Topic human-naming agreement is likewise pending (M2 names are not yet compared). These are stated as not done rather than invented.

5. Combined insights and recommendations

Each recommendation below needs both methods at once and cites at least one time-series and one text-mining number. Every number is read from `reports/review2/`; the machine-readable matrix is `insight_matrix.csv`.

1. Moderate the tags that are still falling fastest, and stop staffing the slower-declining ones.

- *Time series:* ITS post-slope declines are steepest for fastapi -3.89%/wk, reactjs -3.28%/wk, pandas -2.90%/wk; the mildest are excel -1.92%/wk, java -1.74%/wk, android -1.60%/wk (its_results.csv). 6 of 13 tags are classed steep decline by the 26-week forecast.
- *Text mining:* Site-wide frustration share changed -0.2pp after the break (tone_prepost.csv).
- *Action:* Reallocate moderation hours from stabilising tags to the steep-decline tags, and refresh the forecast each quarter.
- *Stakeholder:* Q&A platform product team | *KPI:* Moderation hours per 1,000 questions on steep-decline tags
- *Caveat:* The break is associated with, not caused by, AI adoption; slope ranks are model-based.

2. Turn rising setup, version and migration language into a documentation backlog.

- *Time series:* The tags with rising version/setup keywords are still declining: python -2.34%/wk; javascript -2.46%/wk (its_results.csv).
- *Text mining:* Rising per-tag keywords include python: version (rank 7), toml (rank 19), pyproject (rank 25), pyproject toml (rank 27); javascript: version (rank 16) (keywords_prepost.csv).
- *Action:* Promote these rising version/setup keywords into the DevRel guide backlog for the tags that show them.
- *Stakeholder:* DevRel teams | *KPI:* Guides shipped against the rising-keyword backlog
- *Caveat:* Keyword frequency is a proxy for demand, not a direct measure of documentation need.

3. Route topics that gain share but resolve poorly to an expert queue.

- *Time series:* The Python Execution & Runtime Errors topic rises +1.85pp site-wide (topic_prepost.csv) while python post-slope is -2.34%/wk (its_results.csv).
- *Text mining:* Python Execution & Runtime Errors resolves at 33.2% within 7 days (topic_outcomes.csv).
- *Action:* Send rising, low-resolution topics to an expert routing queue with topic tags.
- *Stakeholder:* developer-forum managers | *KPI:* 7-day resolved rate and time-to-first-answer on routed topics
- *Caveat:* NMF topics are coarse; a question can span several topics.

4. Give first-time askers a guided template where frustration is rising and new accounts resolve less.

- *Time series:* Frustration share rose only for reactjs +0.5pp, openai-api +0.5pp, pandas +0.2pp and fell slightly site-wide (-0.2pp, tone_prepost.csv); python and javascript still decline (-2.34 and -2.46%/wk, its_results.csv).
- *Text mining:* Review 1 SHAP ranks account_age_days second of all features (mean |SHAP| 0.125), so account age matters almost as much as tag volume (shap_top.csv).
- *Action:* Add a guided question template for first posts, prompted when the frustration flag is set.
- *Stakeholder:* Q&A platform product team | *KPI:* 7-day resolved rate for first posts
- *Caveat:* The template is a UX intervention; its effect is not identified by the pre/post analysis.

5. Trigger a review when weekly volume leaves the forecast band.

- *Time series:* The 26-week forecast carries a 95% band; for ALL the week-26 band is 89-201 around 134 (forecasts_26w.csv).
- *Text mining:* Falling topics such as Core Syntax, Arrays & Data Structures signal where traffic is moving (topic_prepost.csv).
- *Action:* Alert the analytics lead when a tag closes a week outside its 95% forecast band.
- *Stakeholder:* Q&A platform product team | *KPI:* Shares flagged and reviewed per quarter
- *Caveat:* Bands are wide for volatile tags, so alerts are a prompt to look, not a conclusion.

6. Treat openai-api as an emerging tag: a step change without a stable trend.

- *Time series:* openai-api shows a +1645% level change but a -2.56%/wk post-slope, with a 33-week half-life and no classifiable 26-week change (its_results.csv, tag_classes.csv).
- *Text mining:* Docker and Android topics also gained share site-wide (topic_prepost.csv), the same emerging-stack pattern.
- *Action:* Monitor openai-api monthly and review routing and tagging once a baseline exists.
- *Stakeholder:* developer-forum managers | *KPI:* Weekly openai-api volume versus its forecast band
- *Caveat:* A single level jump does not imply a lasting slope change.

6. Dashboard guide

The interactive dashboard (Streamlit with Plotly, owned by M1) has seven pages, all responding to shared sidebar filters (tag, date range, period, event markers, linear/log). Pages 2, 3, 4, 5 and 6 read the method outputs directly; page 1 shows the overview and page 7 the question-triage what-if.

Page	Owner	Reads
1. Overview	M1	`ts/series_long.csv`, `data/processed/monthly_tag_tables.csv`
2. Patterns and the 2022 break	M3	`ts/stl_components.csv`, `ts/changepoints.csv`, `ts/its_results.csv`
3. Forecasts	M5	`ts/forecasts_26w.csv`, `ts/backtest_*.csv`, `ts/tag_classes.csv`
4. Topics	M4	`text/topics_top_terms.csv`, `text/topic_monthly_share.csv`, `text/topic_prepost.csv`, `text/topic_outcomes.csv`
5. Tone and complexity	M2	`text/tone_*.csv`, `text/complexity_*.csv`, `text/keywords_prepost.csv`
6. Recommendations	M3	`recommendations.csv`
7. Question triage (stretch)	M5 + M1	saved model, title and error-text inputs

- **Public link:** TODO(M1): public Streamlit URL
- **Screenshots:** TODO(M1): screenshots of the seven dashboard pages

Questions show titles only, each linked to `https://stackoverflow.com/q/<id>`, with a CC BY-SA 4.0 licence notice. No bodies and no user hashes are shown.

7. Limitations

1. **Weights before 2025.** Sampling weights apply only before 2025; later counts are unweighted.
2. **Thirteen tags.** Only the 13 study tags plus ALL are covered, so the picture is not the whole site.
3. **Acceptance is a proxy.** Resolution means an accepted answer within seven days, not a satisfied asker.
4. **VADER is general purpose.** The tone model is a general lexicon with overrides; the frustration flag is exploratory and its validation is pending.
5. **Topics are English prose.** Topic modelling uses cleaned English prose only; code and non-English text are excluded.

6. **Forecasts during a steep decline are uncertain.** SARIMAX 95% intervals covered only 52.0% of endpoints, and 10 fits failed.
7. **Association, not cause.** Every result is associated with or consistent with AI-adoption timing; nothing is caused by it. The pre/post boundary is a comparison date, and the placebo break is included for honesty.
8. **Pending human checks.** Independent tone labels and topic-naming agreement are not yet available and are not claimed.

8. How to reproduce

```
pip install -r requirements.txt
python -m src.ts_patterns --task all         # STL, stationarity, change points, ITS
-> reports/review2/ts/
python -m src.ts_forecast                    # baselines, backtest, forecasts
-> reports/review2/ts/
python -m src.topics                         # topics, keyword shift
-> reports/review2/text/
python -m src.sentiment && python -m src.complexity   # tone and complexity
-> reports/review2/text/
python -m src.review2_m3_insights            # insight_matrix.csv, recommendations.csv
python -m src.review2_final_report           # final_report.md and final_report.pdf
jupyter nbconvert --execute --to notebook --inplace notebooks/07_timeseries.ipynb
jupyter nbconvert --execute --to notebook --inplace notebooks/08_text_mining.ipynb
```

Every heavy step writes to `reports/review2/`; the notebook, dashboard and this report read those files, so a changed CSV changes every downstream number. The final notebook for the submission is `notebooks/review2_main.ipynb` (M1).

9. Contributions and board link

Member	Role	Summary
M1 Abhinav	Dashboard, final notebook, repo and board	`reports/review2/contribution_member1.md` (TODO)
M2 Rishiikesh	Corpus, tone, complexity	`reports/review2/contribution_member2.md`
M3 Akshay	Patterns, ITS, combined insights, final report, deck	`reports/review2/contribution_member3.md` (TODO)
M4 Nishanth	Topics, keyword shift, evaluation	`reports/review2/contribution_member4.md`
M5 Adhikkesh	Forecasts, backtest, tag classes	`reports/review2/contribution_member5.md`

- **Board:** TODO: GitHub project board link
- **Repository:** TODO: repository link
- **Kaggle notebook:** TODO: Kaggle notebook link